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MACHINE LEARNING (25SC2107E)

Y25 - 2026-2027 Trimester - 4

ABSTRACT

Sec No- 1	Team No- 17	Project Title- Waste Segregation Image Classifier

1. Abstract

Improper waste segregation remains a major obstacle to effective recycling and waste management, with a large share of recyclable and hazardous material still being disposed of incorrectly due to manual sorting errors and lack of awareness. This project proposes a Waste Segregation Image Classifier, a deep learning-based system designed to automatically classify waste items into biodegradable, recyclable, and hazardous categories from images, enabling more accurate and scalable sorting than manual methods allow. The dominant approach in prior work has relied on manually engineered sorting rules or simple classifiers trained from scratch, which struggle to generalise across varied lighting, backgrounds, and item appearances. The system is trained on labelled waste image datasets (TrashNet and Kaggle Waste Classification), with images preprocessed and augmented to improve robustness under these varying conditions. Rather than training a convolutional neural network from scratch, the project applies transfer learning using pretrained architectures such as ResNet50 and MobileNetV2, fine-tuning their final layers to adapt to the waste classification task, with a baseline CNN also trained for comparison to evaluate which modelling approach performs best on this dataset. Model performance is assessed using precision, recall, F1-score, and confusion matrices for each waste category, rather than accuracy alone, since certain categories are visually more difficult to distinguish than others. The resulting model is deployed through a simple interface that allows a user to upload or capture an image and receive an instant classification, demonstrating a practical, low-cost pathway toward automated smart-bin systems and highlighting how transfer learning can be applied to a real-world sustainability problem while remaining feasible for classroom-scale computing resources.

Team Details

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